SMART LIBRARY MANAGEMENT SYSTEM

Name: KINTHAN.K.RAJA

USN: 1RUA25BCA0049

Subject: DBMS

SQL CODE:

```
CREATE DATABASE IF NOT EXISTS smart_library;

USE smart_library;

CREATE TABLE IF NOT EXISTS students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT(20) NOT NULL UNIQUE,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    course VARCHAR(100),
    semester INT
);

CREATE TABLE IF NOT EXISTS books (
    id INT AUTO_INCREMENT PRIMARY KEY,
    book_name VARCHAR(200) NOT NULL,
    author VARCHAR(100) NOT NULL,
    category VARCHAR(100),
    available BOOLEAN DEFAULT TRUE
);

CREATE TABLE IF NOT EXISTS loans (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    book_id INT NOT NULL,
    issue_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE,
    status VARCHAR(20) DEFAULT 'Issued',
    FOREIGN KEY (student_id) REFERENCES students(id),
    FOREIGN KEY (book_id) REFERENCES books(id)
);
```

```
CREATE TABLE IF NOT EXISTS reservations (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    book_id INT NOT NULL,
    reservation_date DATE NOT NULL,
    status VARCHAR(20) DEFAULT 'Pending',
    FOREIGN KEY (student_id) REFERENCES students(id),
    FOREIGN KEY (book_id) REFERENCES books(id)
);

CREATE TABLE IF NOT EXISTS fines (
    id INT AUTO_INCREMENT PRIMARY KEY,
    loan_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_status VARCHAR(20) DEFAULT 'Unpaid',
    FOREIGN KEY (loan_id) REFERENCES loans(id)
);
USE smart_library;

SELECT * FROM students;

SELECT * FROM books;

SELECT * FROM loans;

SELECT * FROM reservations;

SELECT * FROM fines;
USE smart_library;

DELETE FROM fines;
DELETE FROM reservations;
DELETE FROM loans;
DELETE FROM books;
DELETE FROM students;
```

MAIN FEATURES

- Add Student – Add new student details to the database.
- View Students – Display the student records stored in MySQL.
- Add Book – Add new books with details such as book name, author, and category.
- View Books – Display the available books and their details.

• Issue Book – Issue a book to a student and store the loan details.

• Return Book – Record the return of an issued book and update its availability.

• Reservations – Reserve books for students.

• Fines – Add and manage fines related to issued books.

DATABASE

Database Name: smart_library

The database contains the following tables:

• Students

• Books

• Loans

• Reservations

• Fines

DATABASE OPERATIONS

INSERT

Used to add new student and book records to the MySQL database.

SELECT

Used to retrieve and display student and book records from the database.

UPDATE

Used to update book availability and loan status when books are issued or returned.

DATABASE CONNECTIVITY

The Java application is connected to MySQL using JDBC. This allows the application to directly store and retrieve data from the database.